Abbas Saleh

Civil Engineering Student, Toronto Metropolitan University

(647) 674-5424 • abbas.k.saleh@gmail.com • linkedin.com/in/abbas-saleh-0a9540368 • abbassaleh.com • Mississauga, ON

Portfolio Summary

Rev 3, 2026-11-09. Full documents at abbassaleh.com.

About

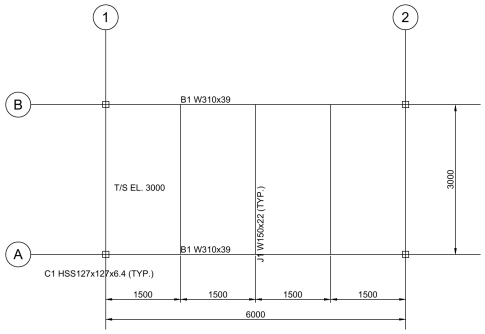
I'm a second-year Civil Engineering student at Toronto Metropolitan University (BEng, expected 2029), focused on structural engineering. I've worked on site with a design-build firm, tying rebar, building formwork and placing concrete, and as a project coordinator on a mixed-use development of two multi-storey buildings with about 182 residential and seniors-living units, coordinating five consulting disciplines (structural, civil, geotechnical, planning, landscape).

Outside class I build verification packages: I check a structure by hand, in a spreadsheet and in analysis software, chase down every difference between them, and issue drawings and models under revision control. My first package has been revised through three rounds of P.Eng. review. Tools: Robot Structural Analysis, Mathcad Prime, Excel, AutoCAD, Revit (Autodesk Certified User); SAP2000 in progress.

Project 1: Maintenance Platform Verification

Self-directed, October 2026. Current revision R3 (2026-10-28). Reviewed by S. Al-Zubaidi, P.Eng. (comments 2026-10-18, 10-21 and 10-28). Next revision Dec 2026.

Structure. 6.0 m × 3.0 m steel platform, top of steel 3.0 m. B1 W310×39 main beams; J1 W150×22 joists at 1.5 m; C1 HSS127×127×6.4 posts; BR1 L64×64×6.4 X-bracing on all four sides, tension-only, bolted at the intersection. W-shapes CSA G40.21 345WM; HSS and angles 350W.
Loads. Dead 0.5 kPa grating + self-weight. Live 4.8 kPa (NBC 2020 Table 4.1.5.3 minimum 3.6 kPa; 4.8 adopted for laydown). Equipment 10 kN at B1 midspan. Guard 0.75 kN/m (NBC 4.1.5.14). Notional 0.005 × 187.3 kN = 0.94 kN, applied as 0.234 kN at each of the four post tops (S16 8.4.1). ULS 1.25D + 1.5L and 1.25D + 1.5L + 1.5G + 1.0N; SLS D + L; deflection L/360.
Method. Three independent checks: Mathcad hand calculation (CAL-001), Excel sheet with named inputs and a CHECK tab (CAL-002), Robot 3D frame model (MOD-001). Lateral loads checked by hand (CAL-003); CSA S16 member checks (CAL-004). Drawn in AutoCAD (S-101, S-201), modelled in Revit (S-301), with an Excel take-off and a review register (MTO-001).



S-101 framing plan (Rev 3).

Gravity results: hand, Excel and Robot agree within 2%

Quantity	Unit	Mathcad	Excel	Robot	Robot vs hand	Limit
J1 factored moment M_f	kN·m	13.51	13.52	13.51	0.0%	
J1 service deflection	mm	3.56	3.56	3.56	0.0%	8.33
B1 factored moment M_f	kN·m	78.70	78.70	78.69	0.0%	
B1 factored shear V_f	kN	35.96	35.96	35.95	0.0%	
B1 service deflection	mm	10.72	10.72	10.72	0.0%	16.67
C1 factored axial load	kN	46.03	46.03	46.83	+1.7%	

The Robot post reaction includes bracing self-weight; the post member force is 46.55 kN (+1.1%). Brace and post forces under the lateral cases agree with Robot within 0.2% (CAL-003). ULS+GY = 1.25D + 1.5L + 1.5GY + 1.0NY governs the posts and braces.

CSA S16 member checks, R3

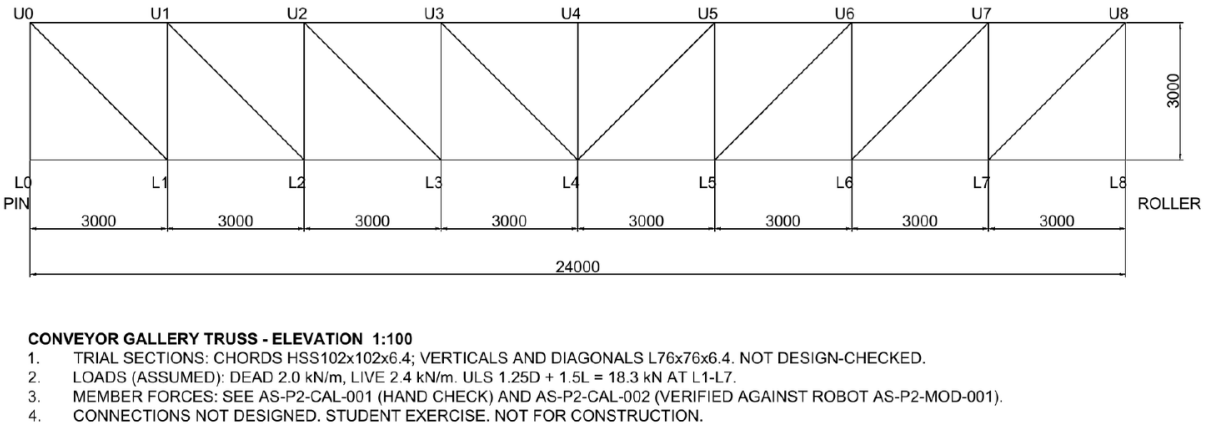
- **B1 governs at 0.99** (biaxial bending, ULS+GY), checked laterally unbraced over 6.0 m with no restraint taken from the joists: $\omega_2 = 1.17$, $M_r = 90.9$ kN·m, strong-axis ratio 78.70/90.9 = 0.87, plus the 0.75 kN/m guard load taken on B1 alone (0.92 with the 1.0 kN point load instead). R1 had assumed restraint at 1.5 m (ratio 0.53); the unbraced check closed that question.
- J1: Class 4 flange in 345WM ($b/2t = 11.5$ against the Class 3 limit $200/\sqrt{F_y} = 10.8$ at 345WM), checked with an effective flange width of 142.1 mm; $M_r = 44.1$ kN·m, ratio 0.31.
- C1: $KL/r = 61.5$, $C_r = 661$ kN, ratio 0.08. BR1: tension and shear lag ≤ 0.05; brace ends 2 × M20 A325, shear 0.03, bearing 0.02.
- Guard torsion: long-edge guard posts moved to the J1 lines and the J1 ends moment-connected to B1 (1.81 kN·m; J1 0.04, bolts 0.04), so B1 carries no torsion. Guard at rail height under 0.9D: minimum post reaction +0.41 kN, no uplift. All other members 0.42 or below.

What the review changed, and what is open

R1 (after the 10-18 comments): braces remodelled as tension-only, which moved the full gravity load to the posts and closed a 10% gap in the post force; notional and guard loads added; S16 checks added. R2 and R3 (after the 10-21 and 10-28 comments): B1 rechecked unbraced, W-shapes rechecked at 345WM, the guard torsion load path fixed, uplift checked, and the brace end connection revised to two bolts. Fifteen items in the review and issue register (eight of them summarized as findings in the brief); I logged my own mistakes in it, including a brace modelled as W6X15 in Revit and caught by the schedule volume check. **Open for Rev 4 (Dec 2026):** remaining reviewer comments, drawing updates (S-201 general note, S-301 title block), publication of the R2/R3 documents, and an independent SAP2000 model of the same frame as a fourth check.

Project 2: Conveyor Gallery Truss, Member Force Verification

Self-directed, October 2026. Rev 0, first issue. Tools: hand calculation, Robot Structural Analysis, Excel, AutoCAD.



Truss elevation, 1:100, sheet S-301 (Rev 0).

24 m	span, 8 panels × 3.0 m, 3.0 m deep, pin and roller supports
73.2 kN	reactions, from the assumed loads worked out by hand
146.4 kN	governing chord force U3–U4 (compression), from the section cut M/d = 439.2 / 3.0
1,317 kg	steel take-off, one truss (1,449 kg with a 10% connection allowance)

What was done. A statically determinate Pratt truss was analysed under assumed loads of 2.0 kN/m dead and 2.4 kN/m live (ULS 18.3 kN at L1–L7). Seven members were solved by hand, by the method of joints at L0, U0 and L1 and a section cut through panel 4, and the hand values matched the Robot member forces once Robot's sign convention (+ = compression) was converted. The elevation was drawn in AutoCAD and a take-off prepared from the CISC section areas, with a member count check ($2j - r = 33$).

What was not done, stated plainly. The remaining 26 members were taken from Robot and checked by symmetry only, which confirms the two halves of the model agree with each other, not that the model is right. The trial sections (HSS102×102×6.4 chords, L76×76×6.4 verticals and diagonals) were not design-checked, and connections were not designed. **Rev 1 (Dec 2026):** hand-solve all 33 members using panel shear for the diagonals, reissue the comparison, and add CSA S16 member checks.

Project 3: Literature Review, Maintenance Access in Hydropower Plants

AS-P3-RPT-001, Rev 0, 2026-10-18. Five pages, six sources.

Question. For turbine maintenance which has to be repeated, what are the practical differences between a permanent modular platform and temporary scaffolding, in fall-hazard exposure, time to get access set up, and the inspection and maintenance which come with each system? The starting point was Hatch's modular access platform for the McCormick hydropower plant (2026); the other sources were Ontario fall-protection regulation (O. Reg. 213/91), provincial working-at-heights guidance, a University of Toronto scaffold standard, one peer-reviewed study and ZARGES product documentation.

Finding. The main benefit of a permanent platform does not come from the inspection itself but from the scaffold work before and after it, which is repeated every outage and exposes workers to fall hazards while the scaffold is incomplete. Worker safety is the strongest case for a permanent platform, but no source quantified the hours saved or incidents avoided, and very little was found on what a permanent platform costs to maintain over its life. This is the weakest piece in the portfolio because it contains no calculation; it is included because it is where the Project 1 platform came from.

Document Register and Status

Document	Description	Rev	Published
AS-P1-RPT-001	Technical brief	R1	Yes (R1)
AS-P1-CAL-001	Mathcad hand calculation	R0	Yes
AS-P1-CAL-002	Excel calculation and three-way check	R0	Yes
AS-P1-CAL-003	Hand check of lateral loads and displacements	R1	Yes (R1)
AS-P1-CAL-004	CSA S16 member checks	R3	R1; R3 on website
AS-P1-MOD-001	Robot 3D model and result plots	R1	Plots
AS-P1-MOD-002, S-301	Revit model and schedule sheet	R0	Yes
S-101, S-201	AutoCAD framing plan, sections and notes	Rev 3	Yes; Rev 4 pending
AS-P1-MTO-001	Material take-off, slenderness, review register	R1	Yes
AS-P2-CAL-001	Truss hand check, 7 members	R0	Yes
AS-P2-CAL-002	Comparison against Robot	R0	Being reissued (Rev 1)
AS-P2-MOD-001, S-301	Robot truss model; AutoCAD elevation (sheet no. to change)	R0	Yes
AS-P2-MTO-001	Truss material take-off	R0	Yes
AS-P3-RPT-001	Literature review	R0	Yes

Package revisions (R0–R3) and drawing revisions (Rev 0–3) are numbered separately. Section properties are from the CISC database; loads marked as assumed were not taken from a cited source. All three projects are self-directed student exercises and are not for construction.

Work planned, December 2026

- Project 1 Rev 4: close the remaining review comments, update the drawings, publish the R2/R3 documents, and rebuild the frame in SAP2000 as an independent fourth check against the hand, Excel and Robot results.
- Project 2 Rev 1: hand solution of all 33 members, reissue of CAL-002, CSA S16 member checks, and renumbering of the elevation sheet, which currently shares the number S-301 with the Project 1 Revit sheet.